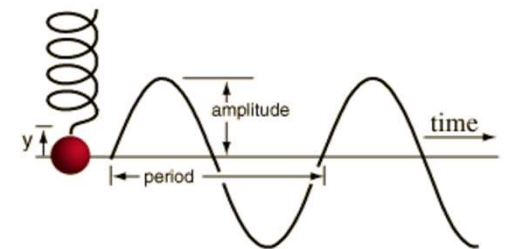
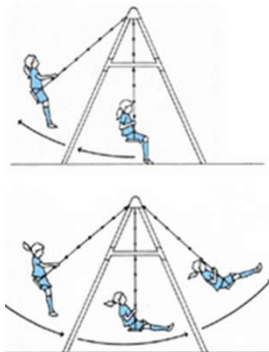


I. Periodic Motion

Repeated motions in equal time intervals, such as those made by a child on a playground swing, or the pendulum of a grandfather clock, Earth in its orbit around the Sun, and wave movement are called a periodic motion.



AMPLITUDE, PERIOD, AND FREQUENCY

amplitude

the maximum displacement from equilibrium

period

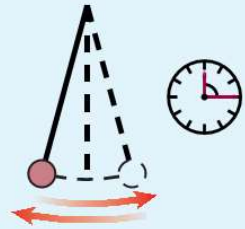
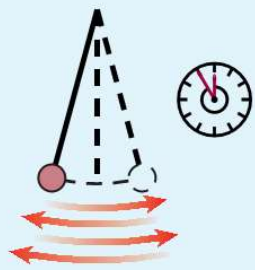
the time that it takes a complete cycle to occur

frequency

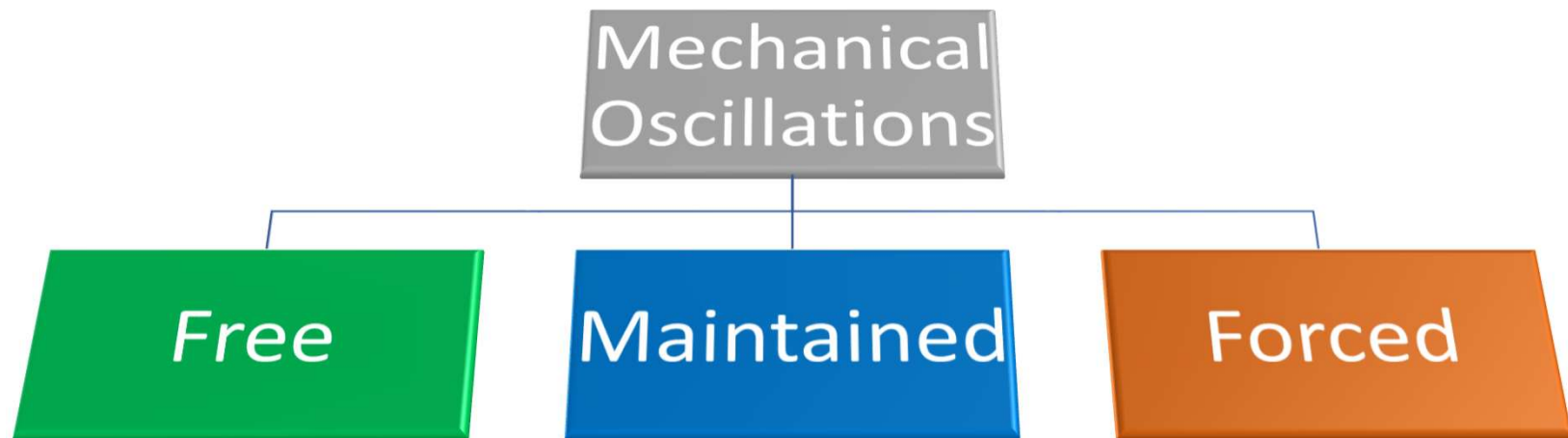
the number of cycles or vibrations per unit of time

$$f = \frac{1}{T} \text{ or } T = \frac{1}{f}$$

Table 2 Measures of Simple Harmonic Motion

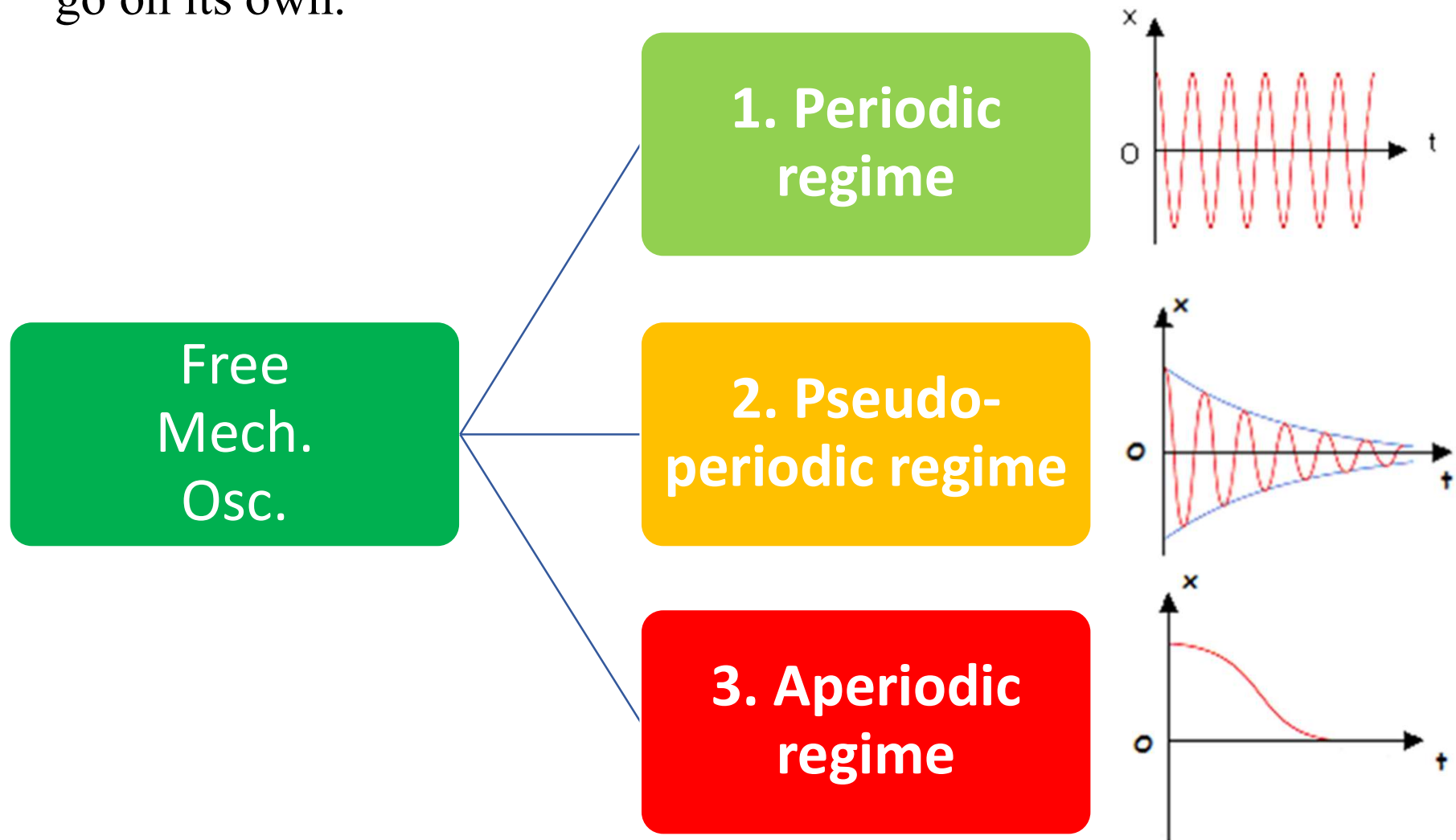
Term	Example	Definition	SI unit
amplitude		maximum displacement from equilibrium	radian, rad meter, m
period, T		time that it takes to complete a full cycle	second, s
frequency, f		number of cycles or vibrations per unit of time	hertz, Hz (Hz = s ⁻¹)

II. Different Modes of Mechanical Oscillations



A. Free Mechanical Oscillations - 3 Regimes

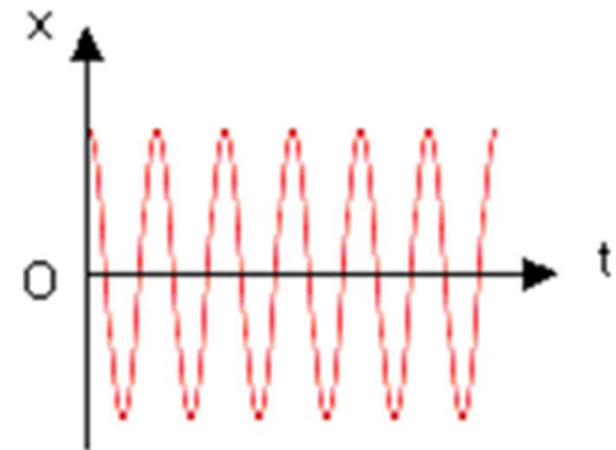
We provide the oscillator with some energy once and then let it go on its own.



1) Undamped oscillations – Periodic Regime

Friction do not exist so there is conservation of mechanical energy:

- $ME = \text{constant}$, thus
$$\frac{dME}{dt} = 0 \quad \forall x \text{ and } t$$
- The amplitude X_m of the oscillations remains constant.
- The regime of the oscillations is said periodic of proper period T_0 .

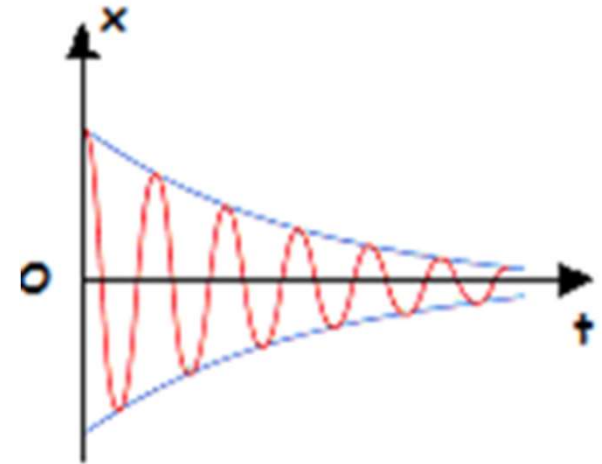


No Friction; the phenomenon is periodic

2) Low damping oscillations – Pseudo-periodic Regime

Friction exists but it is weak (less important):

- Mechanical energy cannot be conserved. It decreases with x & t
- $\Delta ME = W_{\vec{f}}$ thus
$$\frac{dME}{dt} = (W_{\vec{f}})' = \text{Power of } \vec{f}$$
- The amplitude X_m decreases with time t .

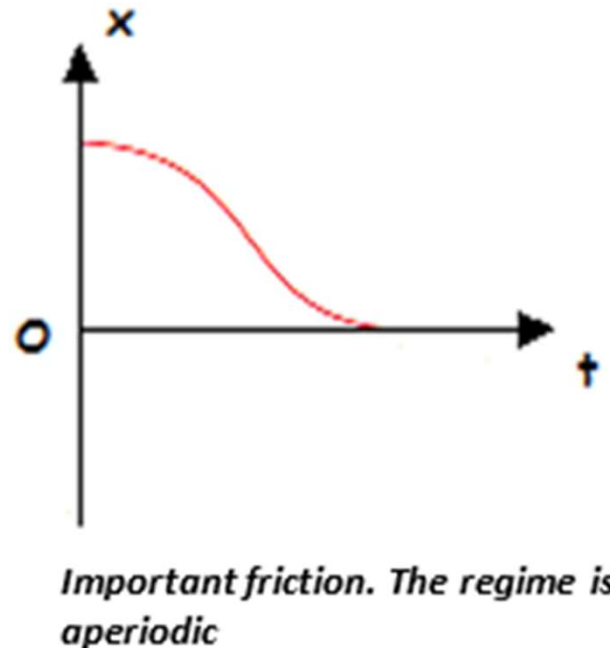


*Low friction; pseudo-periodic regime
of pseudo-period (or false period) T
slightly larger than T_0*

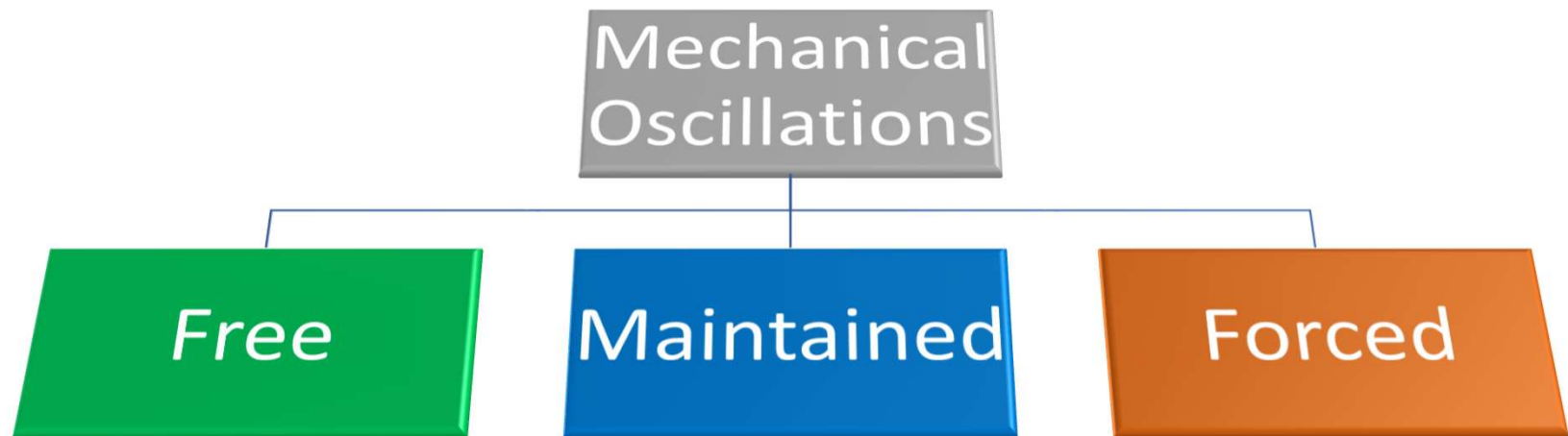
3) Very damped oscillations – Aperiodic Regime

Frictions are very important:

- ME decreases quickly. ME is not conserved.
- The amplitude X_m decreases rapidly with the time t .
- The oscillation regime is called aperiodic.



II. Different Modes of Mechanical Oscillations



B. Maintained mechanical oscillations

- They are mechanical oscillations with little friction.
- They are maintained by supplying the oscillator with energy to compensate the mechanical energy loss.
- Energy supplied to the oscillator during a period is $W = |\Delta E_m|$.
- The mechanical power supplied to the oscillator is:

$$P_{average} = \frac{W}{\Delta t} = \frac{W}{T}$$